# NUMERICAL COMPUTATION OF HADAMARD FINITE PART OF SOME HYPERSINGULAR INTEGRALS IN $1$, $2$ AND $3$ DIMENSIONS

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**ABSTRACT**

Data mining is a good way to find the relationship between raw data and predict the target we want which is also widely used in different field nowadays. In this project, we implement a lots of technology and method in data mining to predict the sale of an item based on its previous sale. We create a strong model to predict the sales. After evaluating this model, we conclude that this model can be used in normal life for future sale's prediction.

## 1 Introduction

Our project is a competition on Kaggle (Predict Future Sales). We are provided with daily historical sales data (including each products' sale date, block ,shop price and amount). And we will use it to forecast the total amount of each product sold next month. Because of the list of shops and products slightly changes every month. We need to create a robust model that can handle such situations.

## 2 Notations

For every physical quantity, the corresponding reference space quantity is denoted by a hat, so that, for example, if $\mathbf{x}$ is a point on a $d$-dimensional variety, then $\hat{\mathbf{x}}$ is the corresponding point in the reference space. If $u$ is a function defined on the variety, then $\hat{u}$ is the corresponding function in the reference space. Furthermore, we denote by $⨍$ the Hadamard finite part of an integral.

## 3 Model presentation

Let's consider an finite part integral of the form

$$I = ⨍_{\hat{\tau}} f(\hat{\mathbf{y}})\mathrm{d}\hat{\mathbf{y}}, \qquad f(\hat{\mathbf{y}}) \underset{\hat{y}\to\hat{x}}{\sim} \frac{1}{||\hat{\mathbf{y}}-\hat{\mathbf{x}}||^{d+1}}. \tag{1}$$

# 4 Injecting Richardson extrapolation into Guiggiani's final formula

## 4.1 Richardson extrapolation

For any angle $\theta$, let's consider a finite sequence of $n$ radial points $\rho_i(\theta)$ such that

$$0 < \rho_1(\theta) < \rho_2(\theta) < \cdots < \rho_n(\theta) < \hat{\rho}(\theta), \tag{2}$$

**Recall**: we can express the two first terms of the Laurent expansion of the complete polar kernel $F(\rho,\theta) = \rho K(\chi(\hat{\mathbf{x}}), \chi(\hat{\mathbf{y}}(\rho,\theta)))J(\hat{\mathbf{y}}(\rho,\theta))\hat{u}(\hat{\mathbf{y}}(\rho,\theta))$ as the following limits

$$F_{-2}(\theta) = \lim_{\rho\to 0} \rho^2 F(\rho,\theta), \qquad F_{-1}(\theta) = \lim_{\rho\to 0} \rho \left( F(\rho,\theta) - \frac{F_{-2}(\theta)}{\rho^2} \right). \tag{3}$$

such that we can write the complete polar kernel as

$$F(\rho,\theta) = \frac{F_{-2}(\theta)}{\rho^2} + \frac{F_{-1}(\theta)}{\rho} + \underset{\rho\to 1}{O}(1) \tag{4}$$

Let $c_{1,i}$ and $c_{2,i}$ be the coefficients used in the Richardson extrapolation of order $n$ to compute the previous limits, so that we can write

$$\boxed{F_{-2}(\theta) = \sum_{i=1}^{n} c_{2,i}\rho_i^2(\theta)F(\rho_i(\theta),\theta)}, \qquad F_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i}\rho_i(\theta) \left( F(\rho_i(\theta),\theta) - \frac{F_{-2}(\theta)}{\rho_i^2(\theta)} \right). \tag{5}$$

By injecting the formula of $F_{-2}(\theta)$ into the formula of $F_{-1}(\theta)$, so that we can write $F_{-1}(\theta)$ only as a linear combination of the complete polar kernel evaluated at the radial points $\rho_i(\theta)$, we obtain

$$F_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i}\rho_i(\theta)F(\rho_i(\theta),\theta) - \sum_{i=1}^{n} c_{1,i}\frac{1}{\rho_i(\theta)} \sum_{j=1}^{n} c_{2,j}\rho_j^2(\theta)F(\rho_j(\theta),\theta).$$

By letting $P_n(\theta) = \sum_{k=1}^{n} \frac{1}{\rho_k(\theta)}$ be a constant that is a polynomial in $\cos\theta$ and $\sin\theta$, we can write the previous formula as

$$\boxed{F_{-1}(\theta) = \sum_{i=1}^{n} F(\rho_i(\theta),\theta)\rho_i(\theta)\Big(c_{1,i} - c_{2,i}P_n(\theta)\rho_i(\theta)\Big)} \tag{6}$$

## 4.2 Final quadrature rule

Let's then denote $I(\theta)$ the integral of the complete polar kernel over the radial direction, defined as:

$$I(\theta) = \int_0^{\hat{\rho}(\theta)} \left\{ F(\rho,\theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \right\} \mathrm{d}\rho + F_{-1}(\theta) \ln \left| \frac{\hat{\rho}(\theta)}{\beta(\theta)} \right| - F_{-2}(\theta) \left\{ \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)} \right\} \tag{7}$$

A final quadrature rule for it is the following:

$$I(\theta) \approx \tilde{I}(\theta) = \sum_{i=1}^{n} \tilde{w}_i(\theta)F(\rho_i(\theta),\theta) \tag{8}$$

With the new weights $\tilde{w}_i(\theta)$ defined as

$$\widetilde{w}_i(\theta) = w_i(\theta) - c_{1,i}\rho_i(\theta)W_{-1} + c_{2,i}\rho_i^2(\theta)\left(P_n(\theta)W_{-1} - W_{-2}(\theta)\right) +$$
$$\rho_i(\theta)\left\{\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right|(c_{1,i} - P_n(\theta)c_{2,i}\rho_i(\theta)) - \left(\frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right)c_{2,i}\rho_i(\theta)\right\}$$

and the auxiliary constants :

- $w_i(\theta) = \frac{\hat{\rho}(\theta)}{2}w_i^{\mathrm{Ref}}$ are the weights of the reference quadrature over the segment $[-1,1]$.
- $W_{-1} = \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i(\theta)} = \sum_{i=1}^n \frac{w_i^{\mathrm{Ref}}}{1+x_i^{\mathrm{Ref}}}$.
- $W_{-2}(\theta) = \sum_{i=1}^n \frac{w_i(\theta)}{\rho_i^2(\theta)} = \frac{2}{\hat{\rho}(\theta)}\sum_{i=1}^n \frac{w_i^{\mathrm{Ref}}}{\left(1+x_i^{\mathrm{Ref}}\right)^2}$.

**Remark**: in the case of Gauss-Legendre quadrature, the constants $W_{-1}$ and $W_{-2}$ can be computed analytically as:

$$W_{-1} = 2H_n, \qquad W_{-2}(\theta) = \frac{2}{\hat{\rho}(\theta)}n(n+1) \tag{9}$$

With $H_n = \sum_{i=1}^n \frac{1}{i}$ the $n$-th harmonic number.

# 5 Why the regularised sum converges: an exact polynomial identity

*(This section was written by Claude, Anthropic, at the author's request. The central identity (13) below was verified numerically.)*

Throughout this section we fix $\theta$ and suppress it from the notation, writing $F_i = F(\rho_i,\theta)$, $w_i$, $\rho_i$, $\hat{\rho}$, etc. We focus on the contribution of the *first line* of the weights $\widetilde{w}_i$ of Section 4.2, namely

$$S_n \;:=\; \sum_{i=1}^n \Big[\, w_i - c_{1,i}\,\rho_i\,W_{-1} + c_{2,i}\,\rho_i^2(P_nW_{-1} - W_{-2})\Big]F_i, \tag{10}$$

which is the discretisation of the regular integral $\int_0^{\hat{\rho}}\{F - F_{-1}/\rho - F_{-2}/\rho^2\}\,\mathrm{d}\rho$ of equation (7). The remaining ($\ln$ and $1/\hat{\rho}$) terms of $\widetilde{w}_i$ merely multiply the finite constants $F_{-1}, F_{-2}$ and raise no difficulty; only $S_n$ does, for the reason recalled now.

## 5.1 The apparent paradox

Let us write $\widehat{F}_{-1}, \widehat{F}_{-2}$ for the *computed* (extrapolated) values, i.e. the right-hand sides of the boxed formulas (5)–(6), to distinguish them from the exact limits (3). Collecting (5)–(6), a direct rearrangement gives

$$S_n \;=\; \sum_{i=1}^n w_iF_i \;-\; \widehat{F}_{-1}\,W_{-1} \;-\; \widehat{F}_{-2}\,W_{-2} \;=\; \sum_{i=1}^n w_i\Big(F_i - \frac{\widehat{F}_{-1}}{\rho_i} - \frac{\widehat{F}_{-2}}{\rho_i^2}\Big) \;=:\; \mathrm{GL}_n[\,h_n\,], \tag{11}$$

the $n$-point Gauss–Legendre rule applied to $h_n(\rho) = F(\rho) - \widehat{F}_{-1}/\rho - \widehat{F}_{-2}/\rho^2$.

By the Remark of Section 4.2, $W_{-1} = 2H_n \to \infty$ and $W_{-2} = \dfrac{2}{\hat{\rho}}n(n+1) \to \infty$. Since $\widehat{F}_{-1}, \widehat{F}_{-2}$ converge to the finite limits $F_{-1}, F_{-2}$, the pieces $\sum_i w_iF_i$ and $\widehat{F}_{-2}W_{-2}$ each diverge *quadratically* in $n$ (both behaving like $F_{-2}\dfrac{2}{\hat{\rho}}n^2$), while $\widehat{F}_{-1}W_{-1}$ diverges *logarithmically*; yet $S_n$ converges. This is the exact arithmetic counterpart of the continuous mechanism: neither $F_{-1}/\rho$ nor $F_{-2}/\rho^2$ is integrable at 0, but their combination with $F$ is. The conjecture that the arithmetic divergences cancel through the weights $w_i$, exactly as the continuous singular terms cancel through $F$, is correct. We prove it below *without ever estimating the divergent pieces* – and, in particular, without invoking any (exponential) convergence rate of the extrapolation.

### 5.2 Extrapolation is interpolation of $G = \rho^2 F$

Set $G(\rho) := \rho^2 F(\rho)$, a regular function with $G(0) = F_{-2}$ and $G'(0) = F_{-1}$ (compare (3)). Let

$$\ell_i(\rho) = \prod_{j \neq i} \frac{\rho - \rho_j}{\rho_i - \rho_j}, \qquad \Pi_{n-1} G = \sum_{i=1}^{n} G(\rho_i)\, \ell_i,$$

denote the Lagrange basis and the degree-$(n-1)$ polynomial interpolating $G$ at the nodes.

**Lemma.** *The Richardson coefficients are the Lagrange weights at the origin, $c_{1,i} = c_{2,i} = \ell_i(0)$, and*

$$\widehat{F}_{-2} = (\Pi_{n-1} G)(0), \qquad \widehat{F}_{-1} = (\Pi_{n-1} G)'(0). \tag{12}$$

*Proof.* Since $\rho_i^2 F_i = G(\rho_i)$, formula (5) reads $\widehat{F}_{-2} = \sum_i \ell_i(0) G(\rho_i) = (\Pi_{n-1} G)(0)$. Logarithmic differentiation of $\ell_i$ gives $\ell_i'(\rho)/\ell_i(\rho) = \sum_{j \neq i} (\rho - \rho_j)^{-1}$, hence at $\rho = 0$,

$$\ell_i'(0) = \ell_i(0) \Big( \frac{1}{\rho_i} - P_n \Big),$$

using $P_n = \sum_k 1/\rho_k$. Evaluating the identities $\sum_i \ell_i \equiv 1$ and $\sum_i \ell_i' \equiv 0$ at 0 yields $\sum_i \ell_i(0)/\rho_i = P_n$. Therefore

$$(\Pi_{n-1} G)'(0) = \sum_i G(\rho_i)\, \ell_i'(0) = \sum_i \rho_i F_i\, \ell_i(0) - P_n\, \widehat{F}_{-2},$$

which is precisely the collapsed formula (6) with $c_{1,i} = c_{2,i} = \ell_i(0)$. □

In words: the construction returns the first two Taylor coefficients at $\rho = 0$ of the interpolating polynomial of $G$.

### 5.3 The exact reduction

**Proposition.** *For $n \geq 3$,*

$$\boxed{S_n = \int_0^{\hat{\rho}} Q_n(\rho)\, \mathrm{d}\rho}, \qquad Q_n(\rho) := \frac{\Pi_{n-1} G(\rho) - \widehat{F}_{-2} - \widehat{F}_{-1}\, \rho}{\rho^2} \in \mathbb{P}_{n-3}. \tag{13}$$

*(For $n \leq 2$, $Q_n \equiv 0$ and $S_n = 0$.)*

*Proof.* By the Lemma, the numerator of $Q_n$ is $\Pi_{n-1} G$ minus its own degree-1 Taylor part at 0; it therefore vanishes to second order at 0, and dividing by $\rho^2$ leaves a polynomial of degree $(n-1) - 2 = n - 3$. At each node, $\Pi_{n-1} G(\rho_i) = G(\rho_i) = \rho_i^2 F_i$, so

$$Q_n(\rho_i) = F_i - \frac{\widehat{F}_{-1}}{\rho_i} - \frac{\widehat{F}_{-2}}{\rho_i^2} = h_n(\rho_i).$$

Hence $S_n = \mathrm{GL}_n[h_n] = \sum_i w_i Q_n(\rho_i) = \mathrm{GL}_n[Q_n]$. Since $\deg Q_n = n - 3 \leq 2n - 1$, the $n$-point Gauss–Legendre rule integrates $Q_n$ exactly, which gives the claim. □

This is the heart of the matter. The regularised sum is, *identically*, the integral over $[0, \hat{\rho}]$ of a bounded polynomial of degree $n - 3$. There is no genuine cancellation catastrophe in exact arithmetic: the divergence of $W_{-1}, W_{-2}$ merely reflects that a single finite number, $\int_0^{\hat{\rho}} Q_n$, has been expressed in a basis whose individual coordinates blow up. In particular $S_n$ is finite and $O(1)$ for every $n$, under no hypothesis on $F$ beyond the existence of the two-term Laurent expansion (4). This is the precise content of the proposed analogy between the discrete and the continuous regularisations.

(The conditioning of *evaluating* $S_n$ in floating-point arithmetic — the growth of $\sum_i |\widetilde{w}_i|$ — is a separate and genuine matter, not addressed here.)

### 5.4 Convergence without analyticity

Let $R := \int_0^{\hat\rho} h_\infty \,\mathrm{d}\rho$ be the exact regular integral, with

$$h_\infty(\rho) = F - \frac{F_{-1}}{\rho} - \frac{F_{-2}}{\rho^2} = \frac{G(\rho) - G(0) - G'(0)\rho}{\rho^2}.$$

Writing $E := \Pi_{n-1}G - G$ for the interpolation error and subtracting from (13),

$$S_n - R = \int_0^{\hat\rho} (Q_n - h_\infty)\,\mathrm{d}\rho = \int_0^{\hat\rho} \frac{E(\rho) - E(0) - E'(0)\rho}{\rho^2}\,\mathrm{d}\rho. \tag{14}$$

The integral form of the Taylor remainder, $E(\rho)-E(0)-E'(0)\rho = \int_0^\rho (\rho-s)\,E''(s)\,\mathrm{d}s$, gives $|E(\rho)-E(0)-E'(0)\rho| \le \frac{\rho^2}{2}\sup_{[0,\rho]}|E''|$, whence

$$\boxed{|S_n - R| \;\le\; \frac{\hat\rho}{2}\,\|(\Pi_{n-1}G)'' - G''\|_{\infty,\,[0,\hat\rho]}\;.} \tag{15}$$

Thus $S_n \to R$ as soon as the interpolant of $G = \rho^2 F$ converges in $C^2[0,\hat\rho]$. For Gauss–Legendre nodes this holds whenever $G$ possesses a fixed finite number of continuous derivatives — by Jackson's theorem on best polynomial approximation, combined with the Markov–Bernstein inequalities controlling the two derivatives against the (slowly growing, $O(\sqrt{n})$) Lebesgue constant of the Legendre nodes. The resulting rate is *algebraic*, set by the smoothness of $G$ alone; analyticity of $F$, and hence the exponential convergence of the underlying Richardson process, is sufficient but in no way necessary.

Two points deserve emphasis. First, the divergent constants $W_{-1}, W_{-2}$ have vanished entirely from the error bound: the regularisation has already been performed exactly, at the algebraic level, by the Proposition, so the cancellation never needs to be estimated. Second, the bound is conservative — the pointwise factor $\sup|E''|$ ignores the cancellation provided by integration against $\mathrm{d}\rho$, so the observed convergence is typically faster. As an illustration, take $G(\rho) = 1 + \rho + \rho^{9/2}$ (so that $F$ is genuinely singular, $F_{-1} = F_{-2} = 1$, and $G \in C^4 \setminus C^5$, hence not analytic) on $[0,1]$: one measures $|S_n - R| = O(n^{-6.7})$ while $W_{-2} = \frac{2}{\hat\rho}n(n+1)$ grows quadratically — a clean numerical confirmation that the apparent divergence is harmless and that the convergence is algebraic, governed purely by the regularity of $\rho^2 F$.

## 6 The final quadrature rule with a single extrapolation coefficient

*(This section, like Section 5, was written by Claude, Anthropic, at the author's request.)*

Leaning on Section 5 — whose Lemma (§5.2) showed that the two families of Richardson coefficients coincide and equal the Lagrange weights at the origin — we record the final quadrature rule in its most compact form. For each $\theta$ define the single coefficient

$$c_i(\theta) \;:=\; c_{1,i}(\theta) = c_{2,i}(\theta) = \ell_i(0) \;=\; \prod_{\substack{j=1\\ j\neq i}}^{n} \frac{\rho_j(\theta)}{\rho_j(\theta) - \rho_i(\theta)}, \tag{16}$$

characterised equivalently by the moment conditions

$$\sum_{i=1}^{n} c_i(\theta)\,\rho_i^k(\theta) = \delta_{k0}, \qquad k = 0, 1, \ldots, n-1, \tag{17}$$

which give in particular $\sum_i c_i = 1$ and $\sum_i c_i/\rho_i = P_n(\theta)$.

This collapse to a *single* coefficient relies on the Laurent expansion (4) proceeding in integer powers of $\rho$ — the gauge $\varphi_j(\rho) = \rho^j$, i.e. $p = 1$ in the notation of the companion note on Richardson extrapolation. Both $F_{-2} = G(0)$ and $F_{-1} = G'(0)$ are then read off the single interpolant $\Pi_{n-1}G$ of $G = \rho^2 F$, which is what forces $c_{1,i} = c_{2,i}$.

With this notation the extrapolated coefficients (5)–(6) read

$$F_{-2}(\theta) = \sum_{i=1}^{n} c_i(\theta)\,\rho_i^2(\theta)\,F_i, \qquad F_{-1}(\theta) = \sum_{i=1}^{n} c_i(\theta)\,F_i\,\rho_i(\theta)(1 - P_n(\theta)\,\rho_i(\theta)), \tag{18}$$

with $F_i = F(\rho_i(\theta), \theta)$. The final quadrature rule is

$$I(\theta) \approx \tilde{I}(\theta) = \sum_{i=1}^{n} \widetilde{w}_i(\theta)\, F(\rho_i(\theta), \theta), \tag{19}$$

where the single coefficient $c_i \rho_i$ now factors out of every correction term:

$$\boxed{\widetilde{w}_i(\theta) = w_i(\theta) + c_i(\theta)\,\rho_i(\theta) \left[ (1 - P_n(\theta)\,\rho_i(\theta)) \left( \ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - W_{-1} \right) - \rho_i(\theta) \left( W_{-2}(\theta) + \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)} \right) \right].} \tag{20}$$

The auxiliary constants $w_i(\theta)$, $W_{-1}$, $W_{-2}(\theta)$ and $P_n(\theta)$ are exactly as in Section 4.2; for Gauss–Legendre nodes, $W_{-1} = 2H_n$ and $W_{-2}(\theta) = \dfrac{2}{\hat{\rho}(\theta)}\, n(n+1)$.

Two structural remarks close the construction. First, the factor $(1 - P_n \rho_i)$ multiplying the logarithmic group is exactly the one already carried by $F_{-1}$ above: the bracket separates cleanly the contribution attached to $\ln|\hat{\rho}/\beta|$ (the regular integral and the logarithmic term) from that attached to the $1/\rho^2$ free term (collected in $W_{-2} + \gamma/\beta^2 + 1/\hat{\rho}$). Second, the purely algebraic part of (20) — the terms in $w_i$, $-W_{-1}$ and $-W_{-2}$, obtained by dropping the $\ln$ and free-term contributions — is precisely the quantity $S_n(\theta)$ of Section 5, hence equals the exact integral of a polynomial of degree $n - 3$; this is what guarantees that (20) stays bounded as $n$ grows despite $W_{-1}, W_{-2} \to \infty$.